

JNTU Hyderabad

REGULAR EXAMINATION ? 2024

CAD/CAM and Automation ? Regular Examination, 2024

Subject Code: ME602 Semester: 6 Regulation: R20

Time: 3 Hours Max. Marks: 70

.....
Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
.....

SECTION A (Unit 1: Geometric Modeling)

1. Define Geometric Modeling. Explain its significance in CAD/CAM and Automation with a suitable example. [12 Marks]
2. How is Geometric Modeling implemented? Give an algorithm / procedure and explain. [12 Marks]
3. List the advantages and limitations of Geometric Modeling. Explain any two in detail. [8 Marks]

SECTION B (Unit 2: Finite Element Analysis Basics)

4. Write short notes on Finite Element Analysis Basics. [12 Marks]
5. Solve a problem related to Finite Element Analysis Basics. Show all the steps clearly. [12 Marks]
6. Solve a problem related to Finite Element Analysis Basics. Show all the steps clearly. [12 Marks]

SECTION C (Unit 3: CNC Programming)

7. Discuss the engineering applications of CNC Programming. [8 Marks]
8. Discuss the engineering applications of CNC Programming. [12 Marks]
9. Compare and contrast the key concepts involved in CNC Programming. [12 Marks]

SECTION D (Unit 4: Computer Aided Manufacturing)

10. Write short notes on Computer Aided Manufacturing. [12 Marks]
11. Derive the expression related to Computer Aided Manufacturing and explain each term. [12 Marks]
12. Compare and contrast the key concepts involved in Computer Aided Manufacturing. [8 Marks]

SECTION E (Unit 5: Industrial Robotics)

13. Compare and contrast the key concepts involved in Industrial Robotics. [12 Marks]
14. How is Industrial Robotics implemented? Give an algorithm / procedure and explain. [12 Marks]
15. Describe the working principle of Industrial Robotics with necessary equations. [12 Marks]

*** End of Question Paper ***